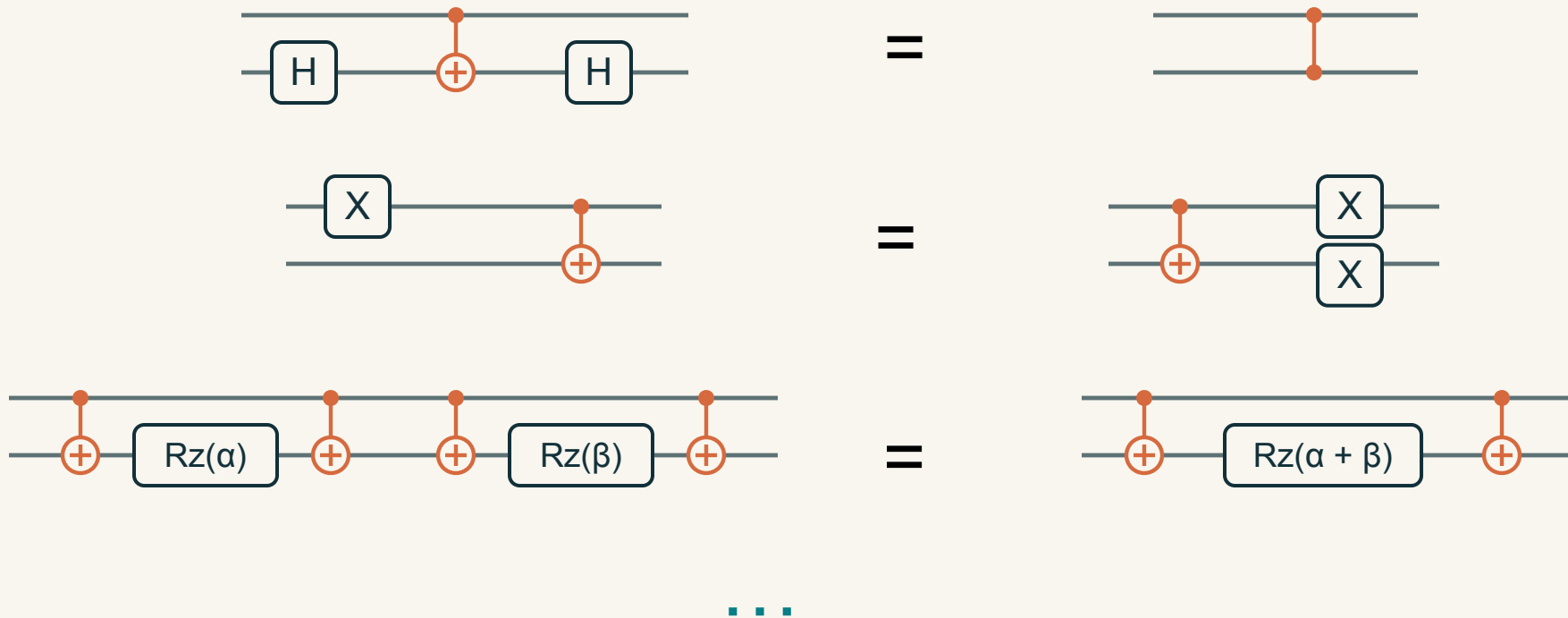


Challenge 2: Large Search Space

Many(infinite) equivalences even for concrete rewrite



Challenge: Hard to provide guaranteed and quantified coverage

What does QSymb do

Goal 1. Symbolic rule that captures all possible patterns of a rewrite

Solution: Capture the full solution of S : **Symbolic constraints** for middle subcircuits S

Goal 2. Small set of rewrite rules with guaranteed coverage

Solution: Filter out redundant rules and propose **canonical symbolic rules** with proven bounded coverage

Goal 3: Generate effective rules for optimization

Solution: Propose **Anchoring** technique that generate effective rules enables for gate reduction at runtime

Results: Outperformed existing work by ~80%!